

Gurkirn Kaur

March 25, 2026

COSC 310 101

Dr. Gema Rodriguez-Perez

Milestone 3 Reflection

Milestone 3 for this project was all about implementing functionality in the backend system. During this milestone I significantly improved my understanding of backend development using the design decisions that were established in Milestone 2. Working with FastAPI helped me learn how endpoints are structured and how different components of a backend system interact. I also got more familiar with organizing code into layers, such as separating routes, services, and data models.

However, the learning process for Milestone 3 was not straightforward. One of the biggest challenges for me was that I had to learn Python almost entirely on my own while also working on this project. Since I did not have a background in Python beforehand, a lot of my time went into understanding syntax, debugging errors, and figuring out how to implement even basic functionality. There were times where I had to close a pull request as my code just wouldn't cooperate, and I would have to fully restart the issue that I was working on. This made it harder to focus on actually building features efficiently and getting a lot of work done. The lab updates did provide some level of structure, but in my experience, they felt more like weekly check-ins rather than opportunities to deepen understanding. They mostly required us to show what we had worked on, rather than helping to clarify difficult concepts. Additionally, there were times when different TAs gave conflicting answers compared to the professor, which made it harder to know which approach was correct. For example, we had to change our branching convention from one branch per feature to one branch for issue, and if we had learned this earlier into the milestone it would have saved time from dealing with merge conflicts and not having multiple people working on the same branch.

The biggest challenge during this milestone for me personally was balancing this project with four other courses, many of which also had large projects. Managing multiple deadlines at the same time was overwhelming, and it was difficult to consistently dedicate enough time to this

project each week. This often resulted in many sleepless nights working under pressure and trying to catch up rather than staying ahead. Working in a team also came with challenges, especially when coordinating work and ensuring consistency across different parts of the codebase. We had to communicate frequently and review each other's work to avoid conflicts and mismatches in implementation. I'm thankful to all my team members since they helped me many times when I was struggling with a feature and continuously having Pylint errors. My group members gave me solid advice on what to do and motivated me to push through. From these experiences, I learned how important it is to manage time effectively and to build a strong foundation in the required tools early on. I also realized that working on large projects requires consistent effort rather than last-minute work.

If I were to redo this phase, I would focus on time management much earlier. I would create a more structured schedule to balance this project with my other courses and avoid falling behind. I would also make it a priority to learn Python before the project started or at least earlier in the semester. Having a stronger foundation would have saved a lot of time and reduced the stress of trying to learn and implement at the same time. Milestone 3 was challenging and at times overwhelming in a way that M1 and M2 were not. I would say I fared much better in those milestones and really pulled my weight more. I aspire to do much more and feel more comfortable for M4 to help my group in the same way they helped me and take on a greater share of work. At the end of the day, this milestone also showed me the importance of preparation, consistency, and managing workload across multiple responsibilities.